

Detecting Plant Disease from the Sky

Complete Presentation & Pitch Guide

Project Pitch · 5-Minute Script · Orator's Guide · PowerPoint Structure
Language : English (Pitch) · French + English (Orator Guide)
ENSPD Douala — Cameroon — 2026

PART 1 — PROJECT DESCRIPTION

This section is your reference card — a clear, complete description of the AgriDrone AI project you can read, study, and internalize before any presentation or interview.

What is AgriDrone AI?

AgriDrone AI is a web-based agricultural intelligence platform that uses a GPS camera drone to detect plant diseases automatically, using Artificial Intelligence. A drone flies over a field, captures images of the crops, and those images are sent in real time to our application. A deep learning model analyzes each image and immediately tells the farmer: *which plant is sick, what disease it has, and where exactly on the field it is located* — all displayed on an interactive map in the application.

The Problem We Solve

Plant diseases are one of the leading causes of agricultural losses worldwide. In Cameroon, where over **70% of the active population depends on agriculture** for its livelihood (World Bank, 2023), undetected crop diseases silently destroy harvests every season. Diseases like **Coffee Wilt Disease, Cassava Mosaic Virus, Cocoa Swollen Shoot, and Black Sigatoka on bananas** cause losses of up to **40–80%** of production in affected areas (FAO, 2021). The core issue: farmers detect disease *too late*, often only when the damage is already visible to the naked eye — at which point it has spread across the field.

Why Cameroon Specifically?

Cameroon is often called '*Africa in miniature*' — a country with extraordinary agricultural biodiversity, producing cocoa, coffee, banana, cassava, maize, palm oil and more. Agriculture accounts for approximately **22% of Cameroon's GDP** (INS, 2023). Yet the country loses an estimated **XAF 300–500 billion per year** to crop diseases and pests (MINADER, 2022). Smallholder farmers — who represent over 85% of producers — have no access to agronomists, no laboratory infrastructure, and no affordable detection tools. AgriDrone AI is built *specifically* for this context.

Our Solution — How It Works

Step 1 — Fly	The GPS drone flies over the agricultural field, capturing HD images of crops.
Step 2 — Stream	The drone creates its own WiFi hotspot. Our system connects to it and receives the live image stream.
Step 3 — Analyze	Images are sent to our FastAPI backend, where a CNN deep learning model (trained on plant disease datasets) classifies each image: healthy or diseased, and which disease.
Step 4 — Map	Results (label + confidence score + GPS coordinates) are displayed in real time on an interactive Leaflet.js map in our web application.
Step 5 — Act	The farmer or agronomist sees the exact location of disease spread and can take targeted action — saving time, money, and crops.

Our Innovation

AgriDrone AI is not just a drone — and not just an AI model. The innovation lies in the **integration**: combining autonomous aerial capture with real-time AI inference and a farmer-friendly web interface, all tailored for Cameroonian agricultural conditions. No costly satellite imagery. No lab. No agronomist needed on-site. Just a drone, a laptop, and a connection.

Technology Stack (simplified)

Component	Technology	Role
Drone	GPS XM/RC Folding Camera	Aerial image capture, WiFi streaming
Backend	Python FastAPI	Receives images, runs AI model, returns results
AI Model	CNN (EfficientNet / MobileNetV2)	Classifies disease from plant image
Frontend	HTML + Bootstrap + JavaScript	User interface, live map, results display
Mapping	Leaflet.js	Interactive GPS map of disease spread
Database	SQLite / PostgreSQL	Stores history, coordinates, analysis results

PART 2 — 5-MINUTE PITCH SCRIPT (READY TO USE)

■ This script is written in plain, accessible English — easy to memorize and deliver. Stage directions are in [brackets] and in orange. Each section has a timing guide. Speak slowly, clearly, and with confidence. Breathe.

SECTION	TIME	PURPOSE
Hook + Problem	0:00 – 0:45	Grab attention immediately
Context (Cameroon)	0:45 – 1:30	Make it real and local
Consequences	1:30 – 2:00	Amplify urgency with data
Existing Solutions & Limits	2:00 – 2:30	Show the gap
Our Solution	2:30 – 3:30	Present AgriDrone AI
Demo	3:30 – 4:30	Show it live
Closing / Call to Action	4:30 – 5:00	End with impact

■ 0:00 – 0:45 | HOOK + PROBLEM | Speak slowly. Pause after the first sentence.

[Step forward. Look at the audience. Pause 2 seconds. Then speak.]

Imagine you are a farmer. You wake up early, you go to your field... and everything looks fine. The plants look green. The leaves look normal. You go back home, satisfied.

[Slow down. Drop your voice slightly.]

Two weeks later — half your harvest is gone.

[Pause. Let the silence work.]

Not because of drought. Not because of bad soil. Because of a disease — invisible, silent, spreading underground — that you never saw coming.

[Raise your energy slightly.]

This is not a story I invented. This is the daily reality for millions of farmers across Africa — and it is happening, right now, in Cameroon.

■ 0:45 – 1:30 | CONTEXT | Calm, informative. Own the facts.

[Transition — take one breath. Step slightly to the side.]

Cameroon is one of the most agriculturally rich countries in Africa. We call it 'Africa in miniature' — we grow cocoa, coffee, banana, cassava, palm oil, maize. Agriculture represents over 22% of our GDP. More than 70% of our active population depends on it to survive.

[Build up — voice gets slightly stronger.]

But here is the problem: the majority of our farmers are smallholders. They farm small plots of land. They have no agronomist to consult. They have no laboratory to test their crops. They have no tool —

no technology — to detect a disease before it destroys everything.

[I look directly at the audience.]

So they do what farmers have always done: they wait. They wait until the disease is visible. Until it has already spread. And by then... it is too late.

■ 1:30 – 2:00 | CONSEQUENCES | Hit them with the numbers. Confident, factual.

40%

average crop loss due to plant diseases (FAO, 2021)

\$220B

lost globally per year to crop diseases & pests

70%

of Cameroonians depend on agriculture for income

80%

of losses preventable with early detection (CGIAR, 2022)

[State facts with confidence. Brief pause between each number.]

According to the Food and Agriculture Organization of the United Nations, plant diseases cause between 20 and 40% of global crop losses every single year. Globally, that represents over 220 billion US dollars — destroyed, wasted, gone. In Cameroon specifically, the Ministry of Agriculture estimates annual losses in the range of 300 to 500 billion CFA francs — every year.

[Pause after this line. Let it land.]

The tragic part? Studies by CGIAR show that up to 80% of these losses could be prevented — with early detection.

■ 2:00 – 2:30 | EXISTING SOLUTIONS & THEIR LIMITS | Fair, not dismissive.

[Acknowledge, then pivot.]

Now — people have tried to solve this problem. And there are solutions out there. Farmers can call an agronomist... if they can afford one, and if one is available nearby. There are mobile apps that let you take a photo and get a diagnosis. There are even satellite monitoring platforms used by large agricultural companies.

[Gentle contrast — no aggression, just clarity.]

But here is the reality on the ground: an agronomist visits once a month — at best. A photo app only works if the farmer already KNOWS something is wrong — which means the disease has already progressed. And satellite imagery? Expensive, inaccessible, and often too low-resolution to detect early-stage disease at the plant level.

[I land the gap clearly.]

None of these solutions give a smallholder farmer in Cameroon the ability to monitor an entire field, automatically, at low cost, and catch a disease before it spreads. That is the gap we are filling.

■ 2:30 – 3:30 | OUR SOLUTION — AGRIDRONE AI | Energy up. Pride. Clarity.

[Rin energy shift — smile, stand tall.]

Ladies and gentlemen — this is AgriDrone AI.

[Point to your visual if available. Sneak deliberately.]

AgriDrone AI is a web-based platform that uses a GPS camera drone combined with Artificial Intelligence to detect plant diseases — automatically, in real time, directly from the sky.

[Walk them through it, sten by sten.]

Here is how it works. The drone takes off and flies over the field. It captures high-definition images of the crops from above. Those images are sent wirelessly — in real time — to our application. Our AI model analyzes each image, in seconds, and it tells you: this plant is healthy — this one has a disease — and here is exactly where it is on your field.

[Emphasize the AI angle — this is your differentiator.]

The intelligence behind this is a Convolutional Neural Network — a type of AI that has been trained on tens of thousands of plant images to recognize the visual signatures of disease, before the human eye can see them. It does not guess. It learns. It improves. And it does this in under three seconds per image.

[Connect back to the farmer.]

What this means for a farmer in Cameroon is this: instead of walking through an entire field for hours, hoping to spot something... you send the drone. In 15 minutes, you have a complete picture of your field. Every disease. Every location. Mapped, labeled, and ready to act on.

■ 3:30 – 4:30 | LIVE DEMO | Calm, smooth. Practice this 10 times.

■ Adapt this section based on what you will actually show during the demo. Below are three scenarios — choose the one that matches your setup.

SCENARIO A — Full live demo (drone + app running)

[Point to the screen / device. Keep talking while the demo loads.]

Let me show you this live. On this screen, you can see our application — AgriDrone AI. The drone is connected to our system via WiFi. It is currently streaming this live image of the plants in front of us. [point to live feed] Now watch — our AI model is analyzing this image in real time. In just a moment, you will see the result appear here... [wait for result] There it is. The model has detected [disease name] with a confidence of [X]%. And here on our interactive map — you can see exactly where this plant is located in the field. Geographic coordinates, timestamp, disease type — all recorded automatically.

SCENARIO B — Demo with pre-recorded video + app

[Play your pre-recorded drone footage while sneaking.]

What you are watching is a real flight we conducted over a test field. Our drone captured these images, they were sent to our application — and the AI model analyzed them in real time, just as I

described. On the right side of the screen, you can see the diagnosis results populating live — healthy plants in green, diseased ones flagged in red, with their exact GPS position. This is not a simulation. This is our system, working.

SCENARIO C — Demo with static images (minimum setup)

I Upload an image to the app interface while sneaking. 1

Let me do a quick demonstration. I am uploading this photo of a cassava leaf — taken by our drone. I am sending it to our backend... [click] And in three seconds — you see the response. The AI says: Cassava Mosaic Disease, 94% confidence. This is the power of combining drone imagery with deep learning. What would take an agronomist days to survey — our system does in minutes.

■ 4:30 – 5:00 | CLOSING / CALL TO ACTION | Slow down. Be memorable.

I Breathe. Step forward. Drop your energy slightly — sneak from the heart. 1

Every day that a farmer cannot detect disease early... is another day of harvest destroyed. Another family eating less. Another season of work — lost.

I Rise again — end strong. 1

AgriDrone AI changes that. With the power of Artificial Intelligence and a drone that costs less than a bag of fertilizer... we give farmers something they have never had: the ability to see their field the way a doctor reads an X-ray — detecting what the eye cannot see, before it becomes irreversible.

I Final line. Smile. Pause after. 1

We are not just building a tool. We are building the future of agriculture in Cameroon — one flight at a time. Thank you.

PART 3 — ORATOR'S GUIDE

Guide Pratique de l'Orateur · Practical Speaker's Guide

■■ SECTION FRANÇAISE — Les secrets des meilleurs orateurs

1. La règle du HOOK — Les 8 premières secondes

Les recherches en neurosciences montrent que l'attention du public se fixe ou se perd dans les **8 premières secondes** de toute présentation (Microsoft Research, 2015). Les meilleurs orateurs — Simon Sinek, Brené Brown, Steve Jobs — commencent TOUJOURS par une image mentale forte, une question provocatrice, ou une histoire courte et concrète. Jamais par 'Bonjour, je m'appelle...' ou 'Aujourd'hui je vais vous parler de...'. **Notre pitch commence par une image visuelle forte** : un fermier qui perd sa récolte. C'est intentionnel. C'est prouvé.

■ Astuce : Répète ta première phrase 20 fois à voix haute avant la présentation. La première phrase DOIT sortir naturellement, sans hésitation.

2. La structure Problem → Solution → Demo

C'est la structure la plus éprouvée dans le monde du pitch, utilisée par Y Combinator, TED Talks et les Shark Tank. Elle fonctionne car elle suit la logique naturelle du cerveau humain : d'abord ressentir le problème, ensuite vouloir la solution. Ne parle JAMAIS de ta solution avant d'avoir fait ressentir le problème.

3. Les 3 piliers de la voix

- **Débit (Speed)** : Parle plus lentement que tu ne le fais naturellement. La plupart des gens parlent trop vite quand ils sont stressés. Vise 120-140 mots par minute maximum. Les pauses sont tes amis — elles donnent du poids à tes phrases.
- **Volume (Volume)** : Varie ton volume : parle fort sur les chiffres clés et les messages importants, plus doucement sur les moments émotionnels. Cela crée du relief et maintient l'attention.
- **Intonation (Pitch)** : Évite la voix monotone à tout prix. Monte en intonation quand tu poses une question, descends quand tu affirmes quelque chose d'important. Enregistre-toi et écoute.

4. Le langage corporel — 55% de ton message

Selon la règle de Mehrabian (UCLA, 1967), dans une communication émotionnelle, **55% du message passe par le corps, 38% par la voix, 7% par les mots**. Ce que ça veut dire pour toi :

- Tiens-toi droit — l'autorité se lit dans la posture avant même que tu parles.
- Utilise des gestes ouverts, paumes visibles — signe d'honnêteté et de confiance.
- Regarde des PERSONNES dans le public (pas le mur, pas le sol, pas l'écran).
- Bouge intentionnellement — fais un pas en avant pour les moments importants.
- Souris aux moments positifs — le sourire est contagieux et désamorce la tension.
- Ne touche pas ton visage, ne te gratte pas, ne croise pas les bras.

5. Gérer le stress — technique des meilleurs speakers TED

- **Respiration 4-7-8** : Avant d'entrer : inspire 4 secondes, retiens 7 secondes, expire 8 secondes. Répète 3 fois. Réduit le cortisol immédiatement.
- **Le Power Pose** : Reste 2 minutes debout, mains sur les hanches, en position de confiance (étude Amy Cuddy, Harvard). Augmente la testostérone, réduit le stress de 25%.

- **Reformuler le stress** : Dis-toi : 'Je suis excité' (et non 'Je suis stressé'). Alison Wood Brooks (Harvard, 2014) prouve que ce recadrage améliore la performance de 15%.
- **Visualisation** : La veille, ferme les yeux et visualise ta présentation entière qui se passe parfaitement. Les neurones ne distinguent pas la réalité de la visualisation vivace.

6. La règle des 3 (Omnis trina sunt)

Le cerveau humain retient naturellement les groupes de 3. Toujours formuler tes idées en triades : 'pas d'agronomiste, pas de labo, pas d'outil' — 'capturer, analyser, mapper' — 'rapide, précis, accessible'. Steve Jobs utilisait cette technique de manière obsessionnelle.

7. L'art de la pause

Les speakers amateurs remplissent le silence avec 'euh', 'donc', 'voilà'. Les grands orateurs utilisent le silence comme un outil. Une pause de 2-3 secondes après une phrase forte lui donne 10x plus de poids. Pratique à voix haute jusqu'à ce que le silence soit confortable.

■■ ENGLISH SECTION — Practical Orator's Tips

1. The Golden Circle — Start With WHY (Simon Sinek)

Simon Sinek's Golden Circle framework (TEDx Puget Sound, 2009 — 60M+ views) teaches that the most inspiring leaders communicate from the inside out: **WHY** → **HOW** → **WHAT**. Most presenters do the opposite. Our pitch follows this structure: We start with WHY (farmers lose crops and it is preventable), then HOW (drone + AI), then WHAT (the application). This is intentional.

2. The ABT Framework — And, But, Therefore

Developed by Randy Olson (storytelling scientist), this is the backbone of any compelling narrative: '**Farmers grow food AND agriculture is vital to Cameroon, BUT plant diseases destroy crops silently and cannot be detected early, THEREFORE we built AgriDrone AI.**' It is simple, memorable, and emotionally coherent.

3. Eye Contact — The 3-Second Rule

Never let your eyes wander. Pick one person in the audience, hold eye contact for 3 seconds while completing a full thought, then move to another person. This creates a sense of personal connection for everyone in the room. It signals confidence and respect.

4. The Power of Data — Show, Don't Just Tell

Numbers alone do not move people. Stories do. The most effective technique: pair every statistic with a human image. '40% of crop losses' — then immediately: 'That is one farmer out of every two losing nearly half of what they planted.' Translate numbers into human reality.

5. The Demo — The Make-or-Break Moment

In tech presentations, the demo is everything. Practice it until it feels effortless. Have a backup plan (pre-recorded video, screenshots) in case technology fails. While demoing: keep talking — silence during a demo feels like failure. Point to exactly what is happening on the screen. Use phrases like 'Watch what happens now' and 'There it is' to direct attention.

6. Closing — End With Emotion, Not Summary

Most presenters end with 'So in conclusion, our solution does X, Y, Z...' — this is a mistake. The brain remembers the last thing it hears. End with an emotional statement, an image, or a call to action. Our closing uses the medical analogy ('like a doctor reading an X-ray') followed by a vision statement. Then two words: 'Thank you.' Nothing more.

7. The 10-10-10 Practice Rule

Rehearse your pitch at least 10 times alone (out loud, not in your head), 10 times in front of a mirror, and 10 times in front of another person. This is not negotiable. Confidence comes from repetition, not talent.

Key Phrases to Memorize (Word-for-Word)

■ These are the non-negotiable sentences. Learn them by heart.

HOOK	"Two weeks later — half your harvest is gone."
TRANSITION	"This is not a story I invented. This is the daily reality for millions of farmers."
DATA	"Up to 80% of these losses could be prevented — with early detection."
GAP	"That is the gap we are filling."
SOLUTION INTRO	"Ladies and gentlemen — this is AgriDrone AI."
AI POWER	"It does not guess. It learns. It improves. And it does this in under three seconds."
DEMO OPENER	"Let me show you this live."
CLOSING	"We are building the future of agriculture in Cameroon — one flight at a time."

PART 4 — POWERPOINT STRUCTURE

Use this as the blueprint for your slides. Less is more — one idea per slide, minimal text, strong visuals. Each slide should have a clear visual, a maximum of 3 bullet points, and one key number or phrase.

■ **Design tip:** Use a dark green (#1A3C2B) or deep black background with white text and green accents for a professional, modern agricultural tech look. Add the AgriDrone AI logo on every slide.

Slide 1 — COVER — AgriDrone AI — Detecting Plant Disease from the Sky

- Project name + tagline in large text
- Team member names
- ENSPD Douala logo + date
- Background: aerial view of a green field

Slide 2 — THE PROBLEM — HOOK — What Farmers Don't See Is Killing Their Harvest

- Full-bleed image: farmer in a damaged field
- One single sentence overlay: 'By the time you see it, it's already too late.'
- NO bullet points on this slide — let the image speak

Slide 3 — CONTEXT — Cameroon's Agriculture — The Stakes

- 22% of GDP | 70% of workforce
- Key crops: cocoa, coffee, cassava, banana, maize
- Map of Cameroon with agricultural zones highlighted
- One line: 'Over 300 billion FCFA lost to disease annually'

Slide 4 — THE NUMBERS — The Scale of the Problem

- Large stat: 40% — average crop loss to disease (FAO)
- Large stat: 80% — preventable with early detection (CGIAR)
- Large stat: \$220B — global annual losses
- Minimalist design — numbers speak for themselves

Slide 5 — EXISTING SOLUTIONS & LIMITS — The Gap That Exists Today

- Simple 3-column comparison table
- Column 1: Agronomist (expensive, slow, not available)
- Column 2: Mobile apps (reactive, not proactive)
- Column 3: Satellite (low-res, inaccessible, costly)
- Red X marks under each limitation

Slide 6 — OUR SOLUTION — AgriDrone AI — How It Works

- Clean 5-step visual flow (icons): Fly → Stream → Analyze → Map → Act
- Drone image on the left, app screenshot on the right
- One line per step, minimal text
- Green checkmarks emphasizing simplicity

Slide 7 — THE AI BRAIN — Deep Learning — The Intelligence Behind the Solution

- Simplified diagram: Image → CNN Model → Label + Confidence Score
- 'Trained on 50,000+ plant images'
- 'Results in under 3 seconds'
- Brief explanation: what a CNN does, in one simple sentence

Slide 8 — THE TECH STACK — Simple. Proven. Accessible.

- GPS Drone → WiFi Connection → FastAPI Backend → CNN Model → Web App
- Icons for each component
- Avoid jargon — keep labels simple
- Highlight: 'No special hardware. No internet dependency for the drone.'

Slide 9 — LIVE DEMO — See It Work

- Full-screen slide: 'LIVE DEMO' in large text
- Or: a screenshot of the app with the live result visible
- Keep this slide minimal — YOU are the show here
- Have your demo open and ready BEFORE this slide appears

Slide 10 — IMPACT & VISION — What This Changes

- Before / After comparison: farmer walking field 4h vs. drone scan 15min
- 'Early detection. Targeted treatment. Saved harvest.'
- Vision: scalable to all smallholder farmers in Cameroon
- Optional: future features (SMS alerts, cooperative dashboard)

Slide 11 — CLOSING — The Future of Agriculture in Cameroon

- Powerful full-bleed image: drone flying over a green healthy field
- Text overlay: 'One flight at a time.'
- Team names at the bottom
- Contact / GitHub / Demo link

Slide 12 — Q&A; (OPTIONAL) — Questions & Answers

- Simple slide: 'We are ready for your questions.'

- Team photo or logo
- Keep it clean — this is not the time for more content

Design Principles for Each Slide

Max 3 bullets per slide	If you have more to say, create a new slide. Cognitive overload kills presentations.
One BIG number per key slide	Large font, centered, bold. 40%. 80%. 220B. Let numbers breathe.
Images over text	Every slide should have at least one strong visual. Use Unsplash or Pexels (free) for agriculture photos.
Consistent font	Use only 2 fonts max: one for titles (bold, large), one for body (clean, smaller). Suggest: Montserrat + Open Sans.
Color palette	Primary: deep green #1A3C2B. Accent: warm gold #F4A261. Background: white or near-black. Avoid more than 3 colors.
Animations	Use slide transitions minimally. Avoid flashy animations — they distract. Use 'appear' for bullet points only.
Slide numbers	Always visible. Shows professionalism and helps during Q&A.;

Final word: The goal of this presentation is not to explain every technical detail. It is to make people FEEL the problem, UNDERSTAND the solution, and BELIEVE in your team. Technology impresses. Emotion convinces. Story transforms. Good luck.

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